NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 25 /03 /2026
Time: 3 Hrs | GTM – 04 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	D	C	D	C	B	C	C	B	C
11	12	13	14	15	16	17	18	19	20
D	C	C	C	D	D	A	B	B	A
21	22	23	24	25					
500	220	25	7	3					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	B	B	B	C	B	A	B	D	D
36	37	38	39	40	41	42	43	44	45
C	A	C	B	C	C	A	C	B	A
46	47	48	49	50					
75	4	25	70	2					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	D	D	B	B	D	A	B	B
61	62	63	64	65	66	67	68	69	70
D	A	B	D	B	A	B	D	A	C
71	72	73	74	75					
73	8	3	2	1					

Rankers Academy JEE

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HINTS
PHYSICS

1. Statement-1 Analysis: This statement defines a perfectly elastic collision. The coefficient of restitution for an elastic collision is $e=1$. This means the relative speed of separation equals the relative speed of approach. Therefore, Statement-1 is True.

   Statement-2 Analysis: In any collision within an isolated system, the total linear momentum is conserved. This applies to elastic, inelastic, and all other types of collisions. Therefore, Statement-2 is True.Explanation Link: The conservation of relative speed (Statement-1) is a consequence of conserving both linear momentum and kinetic energy. Statement-2 only mentions momentum conservation, which is insufficient to explain Statement-1 on its own. Conclusion: Both statements are true, but Statement-2 is not the correct explanation for Statement-1. The correct option is B.

2. $\tau=\frac{mgL}{2}=l\alpha$

   $\frac{ML^2}{3}\alpha=\frac{MgL}{2}$

   $\alpha=\frac{3g}{2L}$

3. The force exerted on the wall is equal to the rate of change of momentum of the water impacting it.

   $F=\frac{\Delta p}{\Delta t}$

   The change in momentum for a mass of water Δm that hits the wall and stops (since it does not rebound) is given by: $\Delta p=p_{final}-p_{initial}=0-(\Delta m)v=-(\Delta m)v$ The magnitude of the force is therefore:

   $F=\frac{(\Delta m)v}{\Delta t}=\left(\frac{\Delta m}{\Delta t}\right)v$

   The term $\frac{\Delta m}{\Delta t}$ represents the mass flow rate. The volume of water hitting the wall per unit time is the product of the cross-sectional area A and the velocity v.

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The mass flow rate is this volume flow rate multiplied by the density of water, ρ.

$$\Rightarrow \frac{\Delta m}{\Delta t} = \rho(Av)$$

Substituting this into the force equation gives:

$$F = (\rho Av)v = \rho Av^2$$

Using the given values: Area, $A = 2\times10^{-3}\ m^2$, velocity, $v = 12\ m\ s^{-1}$ and density of water, $\rho = 1000\ kg\ m^{-3}$

The force is calculated as: $F = (1000)\left(2\times10^{-3}\right)(12)^2$

$$\Rightarrow F = 2\times144 = 288\ N$$

4. The heat developed in a resistor is given by $H = i^2Rt$, where the terms have their usual meaning.

Here, i becomes $2i$ and everything else on the RHS remains the same. So, the heat developed becomes four times.

Hence, the correct answer is $500\times4 = 2000\ J$. Option (D) is correct.

5. Let stone hit the wall at height h

$$h = H - \frac{1}{2}gt^2 \quad .....(i)$$

$$H = \frac{1}{2}g(2t)^2 \quad ....(ii)$$

Solving equation (i) and equation (ii), $h = \frac{3H}{4}$

6. $v_{\parallel}$ to wall remains constant

$v_{\perp}$ to wall becomes $\left(-\frac{1}{2}\right)2\hat{i} = -\hat{i}$

$$\Rightarrow \vec{v}_f = -\hat{i} + 2\hat{j}$$

7. To find the dimensional formula for each physical constant, we use a defining physical equation.

A. For the Boltzmann constant (k_B), we use the equipartition theorem, where the average kinetic energy of a molecule is proportional to temperature, $E \propto k_BT$.

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$$\Rightarrow [k_B] = \frac{[\text{Energy}]}{[\text{Temperature}]} = \frac{[ML^2T^{-2}]}{[K]} = [ML^2T^{-2}K^{-1}]$$

B. For the coefficient of viscosity (η), we use Stokes' law for viscous force, $F = 6\pi\eta rv$.

$$\Rightarrow [\eta] = \frac{[\text{Force}]}{[\text{radius}][\text{velocity}]} = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = \frac{[MLT^{-2}]}{[L^2T^{-1}]} = [ML^{-1}T^{-1}]$$

C. For the Planck constant (h), we use the energy of a photon, $E = h\nu$, where ν is frequency.

$$\Rightarrow [h] = \frac{[\text{Energy}]}{[\text{frequency}]} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

D. For thermal conductivity (K), we use the law of heat

8. $a_{system} = \frac{g}{4}$

For block A not to slip on B, we have $\frac{mg}{4} = \mu mg \Rightarrow \mu = \frac{1}{4}$

9. The required ratio is, $\sqrt{\frac{3.8-0.6}{1.4-0.6}} = \sqrt{\frac{3.2}{0.8}} = 2$

10. The initial potential energy of the system is the sum of the energies of the two individual capacitors:

$$U_i = \frac{1}{2}CV_1^2 + \frac{1}{2}CV_2^2$$

When the capacitors are connected with like plates together, the total charge is conserved and they form a parallel combination. The initial charges are $Q_1 = CV_1$ and $Q_2 = CV_2$. The total charge is $Q_{total} = Q_1 + Q_2 = C(V_1 + V_2)$.

The equivalent capacitance is $C_{eq} = C + C = 2C$.

The common potential, V, across the combination is given by:

$$V = \frac{Q_{total}}{C_{eq}} = \frac{C(V_1 + V_2)}{2C} = \frac{V_1 + V_2}{2}$$

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The final potential energy of the combined system is:

$$U_f = \frac{1}{2}C_{eq}V^2 = \frac{1}{2}(2C)\left(\frac{V_1+V_2}{2}\right)^2 = \frac{C}{4}(V_1+V_2)^2$$

The decrease in electrostatic potential energy, ΔU, is the difference between the initial and final energies:

$$\Delta U = U_i - U_f = \left(\frac{1}{2}CV_1^2 + \frac{1}{2}CV_2^2\right) - \frac{C}{4}(V_1+V_2)^2$$

$$\Rightarrow \Delta U = \frac{C}{4}\left[2V_1^2 + 2V_2^2 - \left(V_1^2 + 2V_1V_2 + V_2^2\right)\right]$$

$$\Rightarrow \Delta U = \frac{C}{4}\left[V_1^2 - 2V_1V_2 + V_2^2\right]$$

$$\Rightarrow \Delta U = \frac{C}{4}(V_1 - V_2)^2$$

11. The two input NAND gates with their inputs tied together function as NOT gates, producing outputs $\overline{A}$ and $\overline{B}$.

These outputs are then fed into the final NAND gate, resulting in the expression $Y = \overline{\left(\overline{A}\cdot\overline{B}\right)}$.

Applying De Morgan's theorem, this expression simplifies to $Y = A + B$, which is the logical OR operation.

12. At midpoint, let speed of particle is v $\frac{1}{2}mv^2 = qE\frac{\pi R}{2}$

$$a_r = \frac{v^2}{R}, a_t = \frac{qE}{m}$$

$$a_{net} = \sqrt{a_t^2 + a_r^2}$$

13. Electric field due to line charge at distance $r, E = \frac{2k\lambda}{r}$

$$\Delta V = -\int_R^{nR} \vec{E}\cdot d\vec{r} = -2k\lambda\int_R^{nR}\frac{dr}{r} = -2k\lambda\ell n(n)$$

Gain in K.E. = loss in P.E.

$$\Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \frac{\lambda Q}{2\pi\varepsilon_0}\ell n(n), \text{ solve for } V$$

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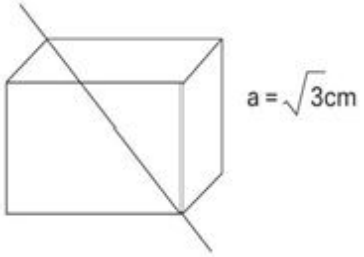
14. For convex lens $\frac{1}{v_1}-\frac{1}{-10}=\frac{1}{20}\Rightarrow v_1=-20$

For plano-convex lens

$$\frac{\mu}{\infty}-\frac{1}{-(d+20)}=\frac{\mu-1}{R}$$

$$\frac{1}{f}=\frac{\mu-1}{R}=\frac{1}{40}\Rightarrow d=20\text{ cm}$$

16.



$$\phi=\frac{q_{enc}}{\epsilon_0}=\frac{\lambda\cdot\sqrt{3}a}{\epsilon_0}$$

$$=2\times10^{-9}\times\sqrt{3}\times\sqrt{3}\times10^{-2}\times36\pi\times10^{9}\,Nm^2C^{-1}$$

$$=2.16\pi Nm^2C^{-1}$$

17. The work done during the cycle ABCA is W = Area of triangle ABC on P – V plane

$$=\frac{1}{2}\times(3V-V)\times(3P-P)=2pV$$

18.

19. $T=2\pi\sqrt{\frac{m}{k}}$

Time period is maximum when K is minimum. In (a), (c) and (d) the spring constants are in parallel so the $K_{eq}=2\,K$. Only in case (b) springs are in series.

So, $K_{eq}=\frac{K}{2}$

Hence time period in this case will be maximum.

20. $\omega=\frac{1}{\sqrt{LC}}$

$i_0=\frac{V_0}{R}$

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21. $\mu_0 ni = B$

$$\mu_0\left(\frac{m}{l}\right)i = B$$

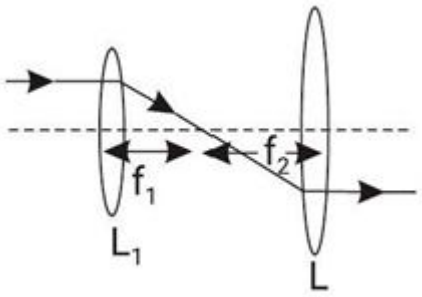
$$m = \frac{Bl}{\mu_0 i} = \frac{6.28\times10^{-3}\times0.5}{12.56\times10^{-7}\times5}$$

$$m = 500$$

22. $\Delta E = (80 \times 7 + 120 \times 8) - (200 \times 6.5)$

$= 220$ MeV

23.



$D = f_1 + f_2 = 25$ cm

Paraxial parallel rays pass through focus and ray from focus of convex lens will become parallel.

24. Area under a-t graph gives change in velocity.

25. If the length, breadth and thickness of a strip are (10.0 ± 0.1)cm, (1.00 ± 0.01)cm and (0.100 ± 0.001)cm, respectively, we have to find the maximum percentage error in its volume.

- The volume is, $V = lbh$, where the terms have their usual meaning.

So, $\ln V = \ln(lbh)$

$$\Rightarrow \ln V = \ln l + \ln b + \ln l$$

$$\Rightarrow \frac{dV}{V} = \frac{dl}{l} + \frac{db}{b} + \frac{dh}{h}$$

(Differentiating)

If the errors are small, we can write dV as ΔV and so on.

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$$\Rightarrow \frac{\Delta V}{V}\times 100 = \frac{\Delta l}{l}\times 100 + \frac{\Delta b}{b}\times 100 + \frac{\Delta h}{h}\times 100$$

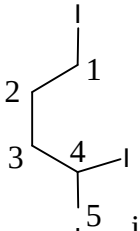
So, we get, $\frac{\Delta V}{V}\times 100 = \frac{0.1}{10}\times 100 + \frac{0.01}{1}\times 100 + \frac{0.001}{0.1}\times 100 = 3$

This $\frac{\Delta V}{V}\times 100$ is the percentage error in V .

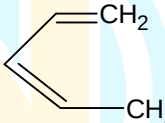
CHEMISTRY

26. $\left[Cu(NH_3)_4\right]^{2+}$ has square planar structure and is paramagnetic.

27. With hot HI



is formed, on elimination gives



, 1,3- Penta diene

28. $t_{1/2} = \frac{0.69}{k} = \frac{0.3\times 2.3}{k}$

$$t_{60\%} = \frac{2.3}{k}\log\frac{100}{100-60} = \frac{2.3}{k}\log\frac{10}{4}$$

$$k = \frac{2.3}{60}\log\frac{10}{4}$$

$$\frac{0.3\times 2.3}{t_{1/2}} = \frac{2.3}{60}\log\frac{10}{4}$$

$$\frac{0.3}{t_{\frac{1}{2}}} = \frac{1}{60}[1-2\log 2] = \frac{1}{60}[1-0.6]$$

$$\frac{0.3}{t_{1/2}} = \frac{0.4}{60} \qquad \therefore\ t_{1/2} = \frac{60\times 0.3}{0.4} = 45\,\text{min}$$

Use direct relation

$$t_{1/2} = 0.3,\ t_{x\%} = \left(\log\frac{100}{100-x}\right)$$

$$t_{60\%} = \log\frac{10}{4} = (0.4)$$

$$\frac{t_{1/2}}{t_{60\%}} = \frac{0.3}{0.4} \qquad \therefore t_{1/2} = t_{60\%}\times\frac{0.3}{0.4} = \frac{60\times 3}{4} = 45\ \text{min}$$

29. Due to the presence of strong field ligands
Fe^{+2} posses 1.72 and Fe^{+3} possess no unpaired electrons

30. NO_2 group, N in the ring and N in diazo (N = N) group do not respond Kjeldahl's test.

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31. $20.6 \text{kcal mol}^{-1}$

$$5C_{(s)} + 4H_2 \rightarrow CH_2 = \overset{CH_3}{\overset{|}{C}} - CH = CH_2$$

$\Delta H = ?$

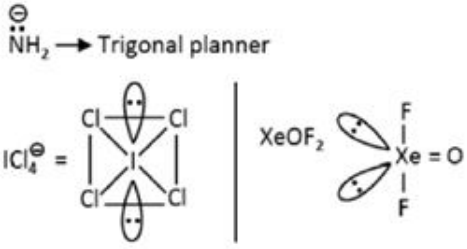
$\Delta H = \Sigma(\text{Bondenergy})_R - \Sigma(\text{Bondenergy})_p$

$= 5 \times C_{(s)-(g)} + 4\Delta H_{H-H} - 2 \times \Delta H_{C=C} - 8H_{C-H} - 2 \times \Delta H_{C-C}$

$\Rightarrow 5 \times 171 + 4 \times 104 - 2 \times 147 - 8 \times 98.8 - 2 \times 83$

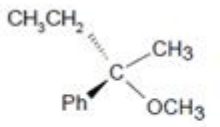
$= 20.6 \text{Kcal mol}^{-1}$

32. NH_2^- Pyramidal



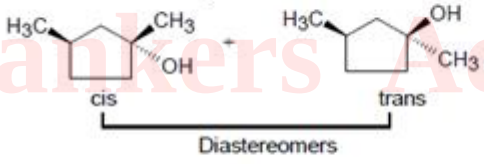
33.

(A)



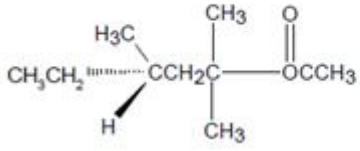
(Racemic), retention and inversion make racemic mixture

(B)



Diastereomers

(C)



(no chiral carbon at the leaving group.)

34.

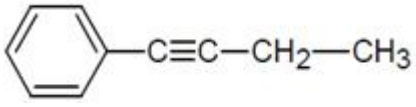
$$H_2C{=}CH_2 + HBr \longrightarrow H_3C{-}CH_2{-}Br$$

(A)

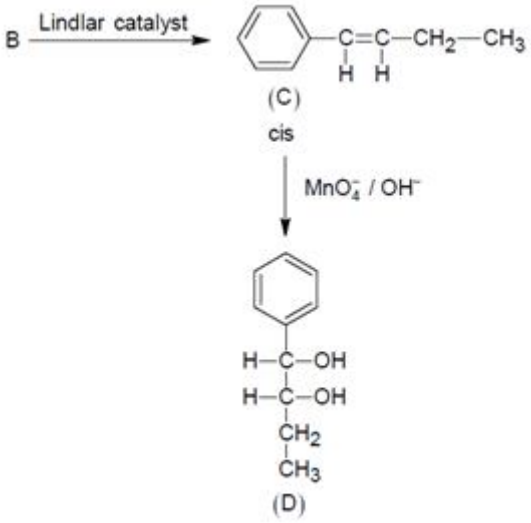
1° alkyl bromide

Thus, (A) is correct.

B is formed by S_N reaction.

B is 

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Lindlar catalyst converts alkyne into cis-isomer. Thus, (C) is correct. dil. MnO_4^- / OH^- and two (- OH) on alkene.

Thus, (B) is correct.

By syn-addtion.

Thus, (C) changes into cis isomer.

Thus, (D) is incorrect.

35. Van't hoof's factor decreases hence freezing point raised.

36. velocity is approximately 6×10^6 m/s.

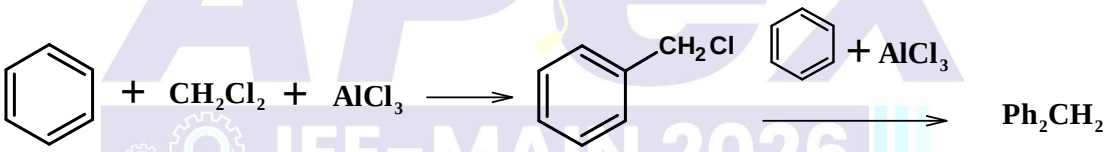
as uncertainty in velocity is very high so we cannot define the trajectroy of an electron.

37. metallic oxides are basic in nature and non-metallic oxides are acidic

38. Na_4XeO_6

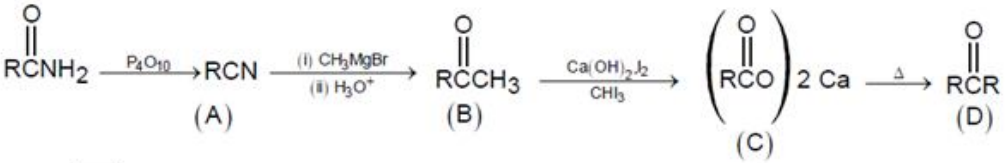
In the compound $NaHXeO_4$, Xenon is in +6 oxidation state. If it disproportionate to give 'Xe' (zero) then Xe (VIII) must be formed. In this reaction $Na_4XeO_6 . 8H_2O$ get precipitated.

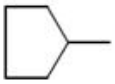
39.

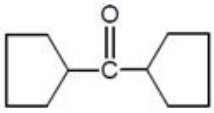


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40.



R is  B gives iodoform test

Thus D is 

41. Less stable intermediate are best hydride donor.

42. $\ln\frac{K_2}{K_1}=\frac{\Delta H}{R}\left[\frac{T_2-T_1}{T_1\,T_2}\right]$

Since, K increases with decrease of temperature and thus, $\Delta H=-ve$.

43. The pH of Zwitter ionic molecules do not depend upon their concentrations.

44. The element is P

45. (A, B, C) Reduction Potential of Cu is higher than that of Zn

46. $\mu_{cal} = q \times d = 1.6 \times 10^{-19}\ C \times 150 \times 10^{-12}\ m = 2.4 \times 10^{-19}\ Cm$

$\mu_{obs} = 1.8 \times 10^{-29}\ Cm$

Percentange ionic character $= \frac{\mu_{cal}}{\mu_{obs}} \times 100 = 75\%$

47. Molecular orbital configuration for up to 14 electrons

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \pi_{2px} < \pi_{2py} < \sigma_{2pz} < \pi^*_{2px} = \pi^*_{2py} < \sigma^*_{2pz}$$

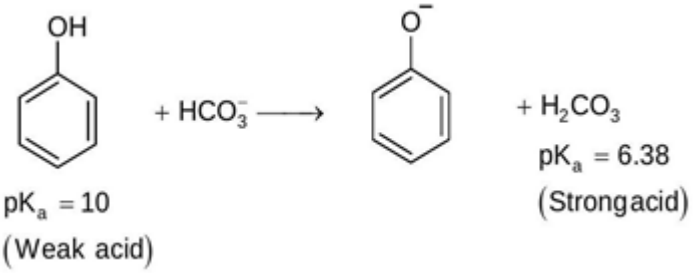
For more than 14 electrons

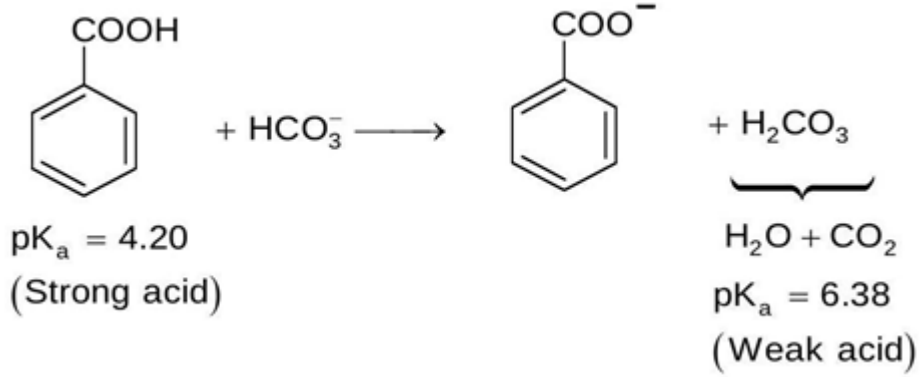
$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi^*_{2px}$$

Number	Species	Total number	No. of electrons in	No. of electrons	Bond order
(i)	O_2^+	15	10	5	2.5
(ii)	CO	14	10	4	3
(iii)	B_2	10	6	4	1.0
(iv)	O_2^-	17	10	7	1.5
(v)	NO^+	14	10	4	3
(vi)	He_2^+	3	2	1	0.5
(vii)	C_2^{2+}	10	6	4	1.0
(viii)	CN^-	14	10	4	3.0
(ix)	N_2	15	10	5	2.5

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48.





With phenol, a strong acid is formed. Thus, equilibrium favours backward reaction.
With benzoic acid, a weak acid is formed, thus, CO_2 is due to it only.

CO_2 formed $= 1$ mol of benzoic acid

$$PV = nRT$$

$$\therefore \quad V = \frac{nRT}{P} = \frac{1 \times 0.0831 \times 300}{1}$$

$$= 24.93 dm^3 \approx 25$$

49. Percentage of yield $x = \frac{92.4}{132} \times 100$

50. $$pH = -\log 5 \times 10^{-10} + \log\left[\frac{5 \times V}{V+10} / \frac{10 \times 2}{V+10}\right]$$

Or $$9 = -\log 5 \times 10^{-10} + \log \frac{V}{4}$$

$$\therefore V = 2 \text{ mL}.$$

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MATHEMATICS

51. We have

$$T=\frac{\tan\alpha(\cos\alpha+1)}{\cot\alpha(\sin\alpha+1)}=\frac{\tan^2\alpha(\cos\alpha+1)}{(\sin\alpha+1)}>0$$

since α is not an integral multiple of $\frac{\pi}{2}$

52. Let $a=n-1,\ b=n, c=n+1$, where $n>1$.

Given $\log(1+ac)=2k$

$\Rightarrow \log(1+(n-1)(n+1))=2k$

$\Rightarrow \log(1+n^2-1)=2k$

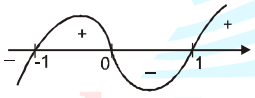
$\Rightarrow \log n^2=2k$

$\Rightarrow 2\log n=2k$

$\Rightarrow \log n=k$

$\Rightarrow \log b=k$

53. $4-x^2\neq 0$, $x^3-x>0 \Rightarrow x\neq \pm 2$ and $-1<x<0$ or $;1<x<\infty$



$\therefore$ D $=(-1, 0)\cup(1,\infty)-\{2\}$ or ; D $=(-1, 0)\cup(1, 2)\cup(2,\infty)$.

54. A ball from first urn can be drawn is two mannars ball is white or ball is black

$$P(w)=\frac{m}{m+n}\quad P(B)=\frac{n}{m+n}$$

Let E $\rightarrow$ selecting a white ball from second urn after a ball from urn first has been placed into it

$$P(E)=P(w)\ P(E/W)+P(B)\ P(E/B)=\frac{m}{m+n}\times\frac{p+1}{p+q+1}+\frac{n}{m+n}\frac{p}{p+q+1}=$$

$$\frac{m(p+1)+np}{(m+n)(p+q+1)}$$

55. Let $f(x)=\frac{1}{a^x-1}+\frac{1}{2}$

Then $f(-x)=\frac{a^x}{1-a^x}+\frac{1}{2}=-f(x)$

Hence $f(x)$ is an odd function. Since so is $F(x)$ and $G(x)=f(x)F(x), G(x)$ is an even function.

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56. $f'(x)=18x^2-90ax+108a^2=0$

$x=2a \;\&\; x=3a$

$x_1=2a \; x_2=3a$

$x_1x_2=54$

$6a^2=54$

$a=3$

$a+x_1+x_2$

$3+2\times3+3\times3=18$

57. Let $3-x^2=t^2$

$+xdx=-tdt$

$f(x)=\int\left(3-t^2\right).t(-tdt)+c$

$=\int\left(t^4-3t^2\right)dt+c$

$=\frac{t^2}{5}-t^3+c$

$f(x)=\frac{\left(3-x^2\right)^{5/2}}{2}-\left(3-x^2\right)^{3/2}+c$

$f\left(\sqrt{2}\right)=\frac{1}{5}-1+c=-\frac{4}{5}$

$c=0$

$f(1)=\frac{2^{5/2}}{5}-2^{3/2}$

$=2^{1/2}\left(\frac{4}{5}-2\right)$

$f(1)-\frac{-6\sqrt{2}}{5}$

58. The given word is MATHS
Arranging the letters alphabetically, we get AHMST
When the word starts with any of the letters A/H/M/S, the number of possibilities
= 4! x 4 =96
Now when the word starts with TA, then the number of possibilities = 3! =6
Now when the word starts with THAMS, then the number of possibilities =1
Rank = 96+ 6+ 1 =103
Hence, rank of the work THAMS is 103.

59. $\tan\alpha+\cot\alpha=a \Rightarrow \tan^2\alpha+\cot^2\alpha+2=a^2$

$\tan^4\alpha+\cot^4\alpha=(a^2-2)^2-2=a^4-4a^2+2$

60. We know that, $5^4=625=52\times12+1$

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$\Rightarrow 5^4 = 52\lambda + 1$, where λ is a positive integer.

$\Rightarrow \left(5^4\right)^{24} = (52\lambda+1)^{24}$

$= {}^{24}C_0(52\lambda)^{24} + {}^{24}C_1(52\lambda)^{23} + {}^{24}C_2(52\lambda)^{22} + \ldots + {}^{24}C_{23}(52\lambda) + {}^{24}C_{24}$ (by binomial theorem)

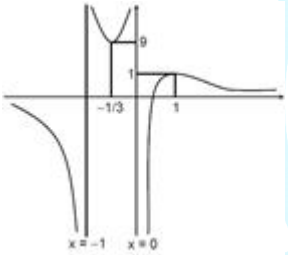
$\Rightarrow 5^{96} = 52\left[{}^{24}C_0 52^{23}\lambda^{24} + {}^{24}C_1 52^{23}\lambda^{22} + \ldots + {}^{24}C_{23}\lambda\right] + 1$

$= \left(\text{a multiple of} 52\right) + 1$

On multiplying both sides by 5 , we get

61. $f(x) = \dfrac{3x-1}{x^2+x}, x \neq -1, 0$

$f'(x) = \dfrac{-(3x+1)(x-1)}{\left(x^2+x\right)^2}$



62. (i) $f'(x) = 5^{g(x)} \ln 5 \; g'(x)$

Now vc $g'(x) = \dfrac{x^2 \cdot 2x}{\ell n\ (1+x^4)} \therefore f'(x) = 5^{g(x)} \ln 5 \dfrac{2x^3}{\ell n\ (1+x^4)}$

$\Rightarrow f'(\sqrt{2}) = 5^{g(\sqrt{2})} \ln 5 \dfrac{2(\sqrt{2})^3}{\ell n(1+(\sqrt{2})^4)} = 4\sqrt{2}$

63. Given,

$n_1 = 10, \bar{x}_1 = 30, CV_1 = 50$

$n_2 = 20, \bar{x}_1 = 25, CV_2 = 60$

$CV_1 = \dfrac{\sigma_1}{\bar{x}_1} \times 100 \Rightarrow 50 = \dfrac{\sigma_1}{30} \times 100 \Rightarrow \sigma_1 = 15$

$CV_2 = \dfrac{\sigma_2}{\bar{x}_2} \times 100 \Rightarrow 60 = \dfrac{\sigma_1}{25} \times 100 \Rightarrow \sigma_2 = 15$

Combined mean $= \dfrac{10 \times 30 + 20 \times 25}{30} = \dfrac{80}{3}$

$d_1 = \bar{x}_1 - \bar{x} = \dfrac{10}{3}, d_2 = \bar{x} - \bar{x}_2 = \dfrac{5}{3}$

Combined variance $= \dfrac{n_1\sigma_1^2 + n_2\sigma_2^2 + n_1d_1^2 + n_2d_2^2}{n_1 + n_2}$

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$$= \frac{2250 + 4500 + \frac{1000}{9} + \frac{500}{9}}{30} = \frac{6225}{27} = \frac{2075}{9}$$

64. For skew symmetric matrix

$a_{ij} = -a_{ji}$

$\Rightarrow$ Only 6 position are to be filled

$\Rightarrow$ Number of ways $= 7^6$

65. $\lim\limits_{x \to 2^-} f(x) = \frac{3}{5} = f(2) \neq \lim\limits_{x \to 2^+} f(x) = 1$

$f(x)$ is not continous at $x = 2$

$\lim\limits_{x \to 3^-} f(x) = \lim\limits_{x \to 3^+} f(x) = f(3) = \frac{9}{2}$

Now LHD ($x = 3$) is

$$\lim_{h \to 0} \frac{\frac{1}{4}((3-h)^3 - (3-h)^2) - \frac{9}{2}}{-h} = \lim_{h \to 0} \frac{h^2 - 8\ \ h + 21}{4} = \frac{21}{4}$$

and RHD ($x = 3$) is $\lim\limits_{h \to 0} \frac{\frac{9}{4}(|h-1| + |-1-h|) - \frac{9}{2}}{h} = 0$

66. By applying row operation in the given matrices and comparing them we can obtain the required result.

(a) The augmented matrix is given by $(A \mid B) = \left(\begin{array}{cccc|c} 3 & -2 & 1 & b & \\ 5 & -8 & 9 & 3 & \\ 2 & 1 & a & -1 & \end{array}\right)$

Applying $R_1 \to 2R_1 - R_2$, we get $(A \mid B) \sim \left(\begin{array}{cccc|c} 1 & 4 & -7 & 2b-3 & \\ 5 & -8 & 9 & 3 & \\ 2 & 1 & a & -1 & \end{array}\right)$

Applying $R_2 \to R_2 - 5R_1, R_3 \to R_3 - 2R_1$, we get

$$(A \mid B) \sim \begin{pmatrix} 1 & 4 & -7 & 2b-3 \\ 0 & -28 & 44 & 18-10b \\ 0 & -7 & a+14 & 5-4b \end{pmatrix}$$

The system of equations will have no solution if $\frac{-28}{-7} = \frac{44}{a+14} \neq \frac{18-10b}{5-4b}$

$\Rightarrow a + 14 = 11$ and $20 - 16\ b \neq 18 - 10\ b$

$\Rightarrow a = -3$ and $b \neq -1/3$.

67. $\cos\beta\pi = \frac{1}{2} \Rightarrow \beta = \pm\frac{1}{3} + 2n, n \in I$

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$$\lim_{x\to\pi}\frac{\sin\alpha x}{1-2\cos\beta x}=-\sqrt{3}$$

$$\Rightarrow\frac{(-1)^{\alpha}\cdot\alpha}{2\beta\cdot\frac{\sqrt{3}}{2}}=-\sqrt{3}\Rightarrow\alpha=1\ \beta=\frac{1}{3}\ \text{for}\ (\alpha\beta)\ \text{minimum.}$$

68. $f'(x)=3x^2+2ax+b$

for $f(x)$ to be increasing $4(a^2-3b)\le 0$

Req. probability $=\frac{16}{36}=\frac{4}{9}$

69. $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}=\frac{1-e}{1+e}$ or $\frac{1+e}{1-e}$

70. Let feet of perpendicular is $(2\alpha, 3\alpha+2, 4\alpha+3)$

$(2\alpha, 3\alpha+2, 4\alpha+3)$

D' ratio of the perpendicular < line < $2\alpha-3, 3\alpha+3, 4\alpha-8$ >

< $2\alpha-3, 3\alpha+3, 4\alpha-8$ > and D' ratio of the line < 2, 3, 4 >

< 2, 3, 4 > $2(2\alpha-3)+3(3\alpha+3)+4(4\alpha-8)=0$

$29\alpha-29=0$

$\alpha=1$

feet of perpendicular is (2, 5, 7)

length perpendicular is $\sqrt{1^2+6^2+4^2}=\sqrt{53}$

71. $T=S_1$

$$\frac{3x}{25}+\frac{y.1}{16}-1=\frac{9}{25}+\frac{1}{16}-1$$

$$\frac{3x}{25}+\frac{y}{16}=\frac{169}{400}$$

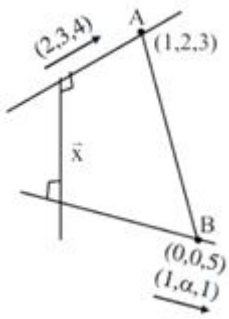
$48x+25y=169$

72. $(x-1)^4=16$

$x=3,-1+2i,1-2i$

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73.



$$L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

$$L_1 : \frac{x}{1} = \frac{y}{\alpha} = \frac{z-5}{1}$$

$$\vec{x} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & \alpha & 1 \end{vmatrix} = \hat{i}(3-4\alpha) - \hat{j}(-2) + \hat{k}(2\alpha-3)$$

$$\text{S.D.} = \left|\frac{\overrightarrow{BA}\cdot\vec{n}}{|\vec{n}|}\right| = \left|\frac{\left(\hat{i}+2\hat{j}-2\hat{k}\right)\cdot\vec{n}}{|\vec{n}|}\right|$$

$$\Rightarrow 6(13-8\alpha)^2 = 25\left((4\alpha-3)^2+(2\alpha-3)^2+16\right)$$

$$6\left(64a^2-280\alpha+169\right) = 25\left(20\alpha^2-36\alpha+34\right)$$

$$\Rightarrow 116\alpha^2+348\alpha-164=0$$

$$\alpha_1+\alpha_2 = \frac{-348}{116} = -3$$

74. $$\lim_{x\to\infty}\sum_{r=1}^{n}\left(\tan^{-1}\left(r+\frac{1}{2}\right)-\tan^{-1}\left(r-\frac{1}{2}\right)\right)$$

$$\Rightarrow \frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right) = \cot^{-1}\left(\frac{1}{2}\right) = \tan^{-1}2$$

75. $$\begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}^2$$

$$= (1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$$

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